

House Pricing Classification

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Introduction

The real estate market plays a significant role in any country's economy, and accurate property price estimation is crucial for buyers, sellers, agents, and investors.

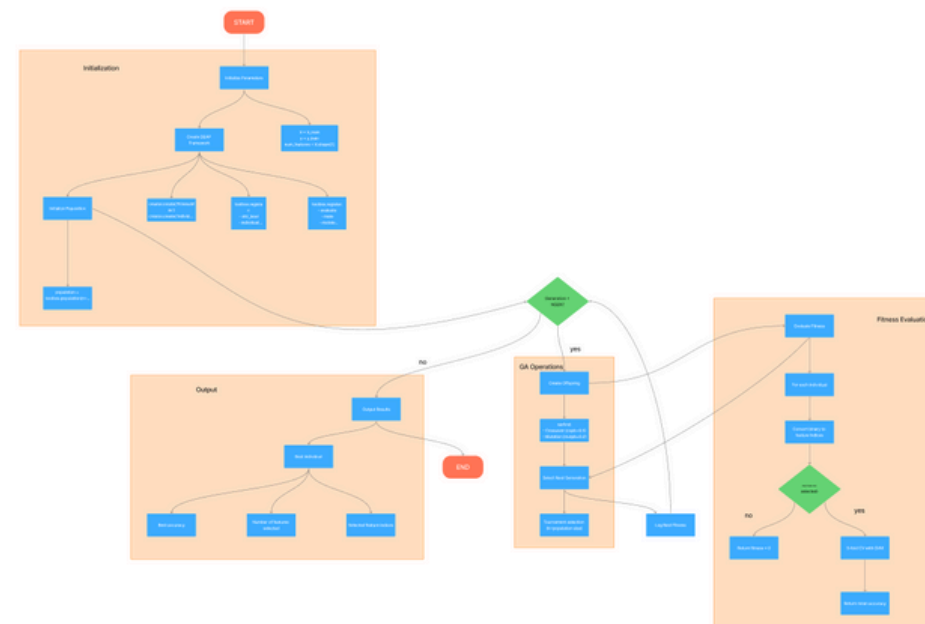
DataSet

The dataset includes various features related to the houses such as size, location, condition, and other attributes that influence the sale price, and have 81 Feature.

Methodology

Data Collection and Preparation

Gathered dataset containing various building features such as location, size, number of rooms, age, and other relevant attributes. Performed data cleaning and preprocessing, including handling missing values and normalizing features. Feature Selection Selected the most important features influencing building prices to improve model accuracy and reduce complexity. Model Training Applied three machine learning models: XGBoost, Random Forest, Support Vector Machine (SVM). Each model was trained on the processed dataset to learn the relationship between features and building prices.



Results

After applying the models, fitting and evaluating the models, we notice that our Accuracy are:

SVM before and after GA: 86% - 83%

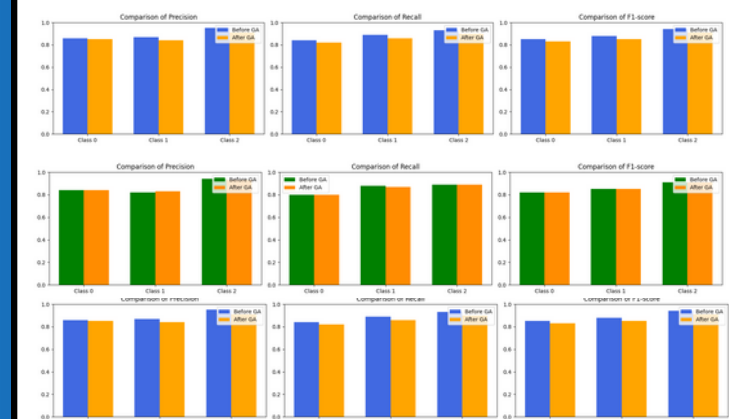
RF before and after GA: 86% - 86%

XGboost before and after GA: 88% - 86%

SVM before and after PSO: 86% - 88%

RF before and after PSO: 86% - 88%

XGboost before and after PSO: 88.9% - 88%



References

- [1] Kaggle : <https://www.kaggle.com/datasets/prevek18/ames-housing-dataset>
- [2] wikaspedia: <https://en.wikipedia.org/wiki/XGBoost>
- [3] wikipedia: https://en.wikipedia.org/wiki/Random_forest
- [4] wikipedia: <https://en.wikipedia.org/wiki/SVM>
- [5] GitHub: https://github.com/MAMDOUHjr/Soft_computing_project/blob/main/Soft_final_Project.ipynb
- [6] Dataset: <https://www.kaggle.com/datasets/asadullahgalib/guava-disease-dataset/data>